

2023 AMC 10B Problem 14 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 14

Problem:

How many ordered pairs of integers (m, n) satisfy the equation $m^2 + mn + n^2 = m^2n^2$?

Answer Choices:

- A. 7
- B. 1
- C. 3
- D. 6
- E. 5

Solution:

The equation may be rewritten as $(m + n)^2 = mn(mn + 1)$. Integers mn and $mn + 1$ are consecutive, so their product is a square of an integer if and only if one of them is 0. Thus $mn = -1$ or $mn = 0$, and the ordered pairs $(0, 0)$, $(-1, 1)$, and $(1, -1)$ are the only (C) 3 ordered pairs of integers that satisfy the given equation.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) [↗](#)

